6 Respiratory system

Section A

[2010-2]

Figure 2 shows the respiratory structure of an insect. Use it answer the questions that follow.

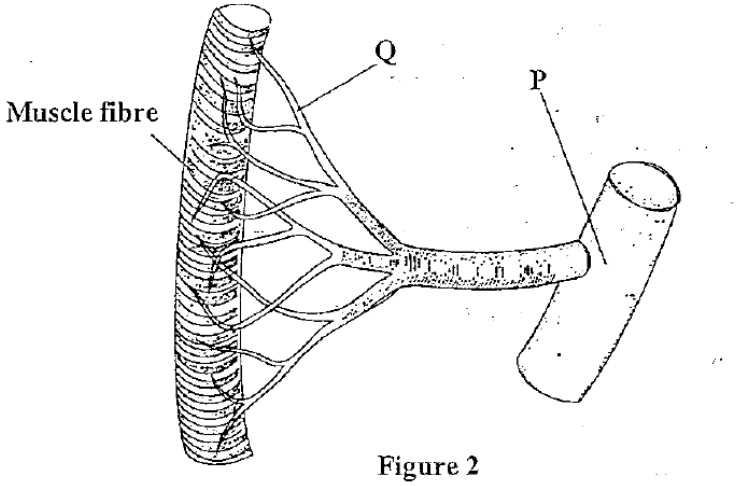

Figure 2

a. Name the part marked **P**. (1 mark)

b. State one adaptation of the part marked **Q** to its function. (1 mark)

c. Explain how oxygen moves from part marked **Q** to the muscle fibre. (2 marks)

[2009-2]

Figure 1 shows part of the lining of the trachea. Use it to answer the questions that follow.

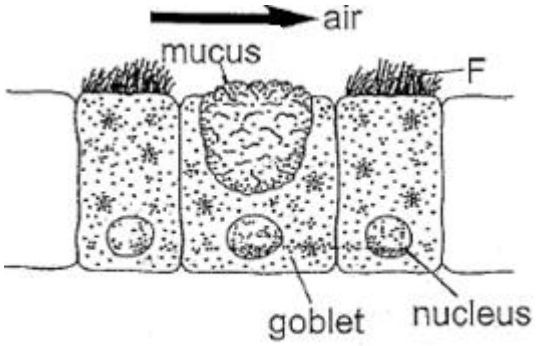

Figure 1

2009-2.

a. Name the part marked **F**. (1 mark)

b. What is the function of each of the following?

(i) Mucus released by the goblet cell (1 mark)

(ii) Pleural fluid (1 mark)

c. Explain how the trachea is kept open during breathing. (2 marks)

Section B

2012-10.

a. Mention any **two** effects of smoking on human health. (2 marks)

b. State any **two** effects of exercise on breathing. (2 marks)

c. Mention any **two** adaptations of alveoli to their function. (2 marks)

[2011-13]

Figure 10 is a graph showing the population of the bacteria and amount of oxygen in a stream after sewage is discharged into it. Use it to answer the questions that follow.

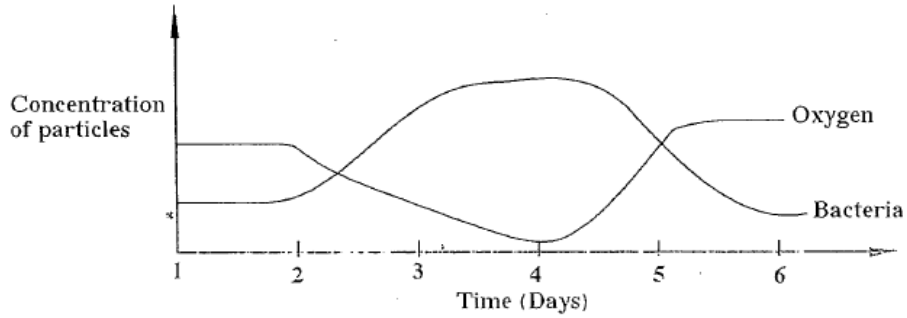

Figure 10

2011-13.

a. When was sewage discharged into the stream?(1mark)

b.

(i)What happened to the amount of oxygen between day 2 and day 4? (1mark)

(ii)Explain your answer given in 13.b.(i). (2marks)

[2009-12]

Figure 10 is a diagram of an alveolus surrounded by a blood capillary. Use it to answer the questions that follow.

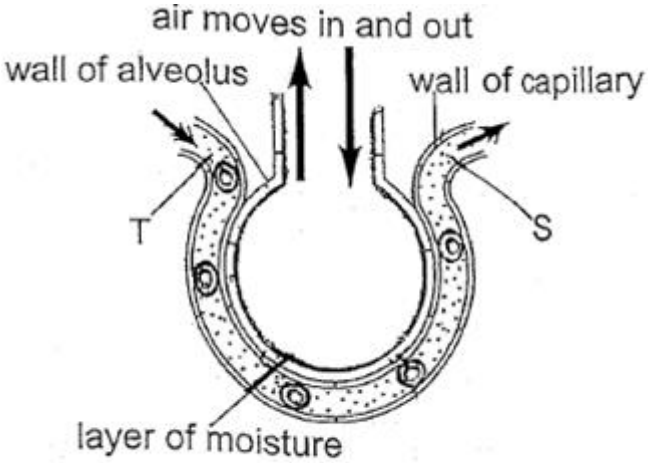

Figure 10

2009-12.

a. To which chamber of the heart does blood flow from point **S**? (1 mark)

b. Explain why there is a difference in the amounts of oxygen in the blood between points **T** and **S**. (2 marks)

c. Explain the role of the thin layer of moisture in the alveolus. (2 marks)

[2007-7]

Figure 5 shows volume of air in the lungs of a child at rest to the start of an exercise. Use it to answer the questions that follow.

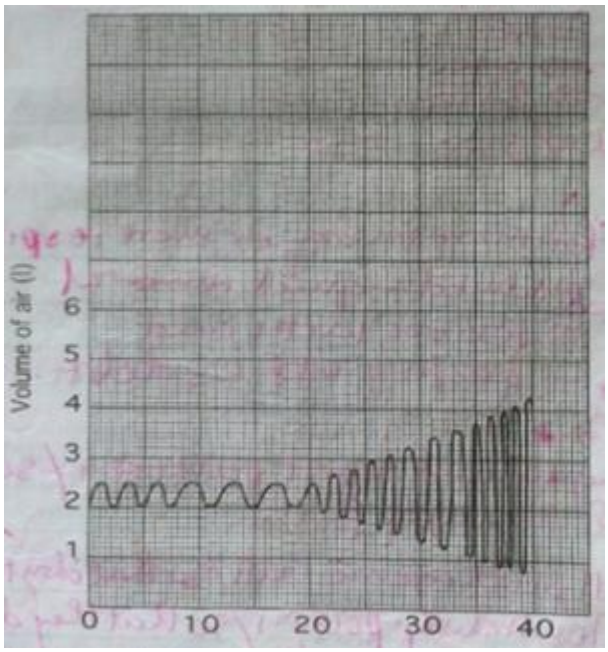

Figure 5

2007-7.

a. At what time does the child start the exercise? (1 mark)

b. What is the maximum volume of air inspired during the exercise? (1 mark)

c. Calculate the number of breaths per minute. (4 marks)

(i) at rest

(ii) after exercise

d. Explain why there is an increased breathing rate between 20 – 40 seconds? (2 marks)

[2005-9]

Figure 4 is a diagram showing a summary of the process of respiration. Use it to answer the questions that follow.

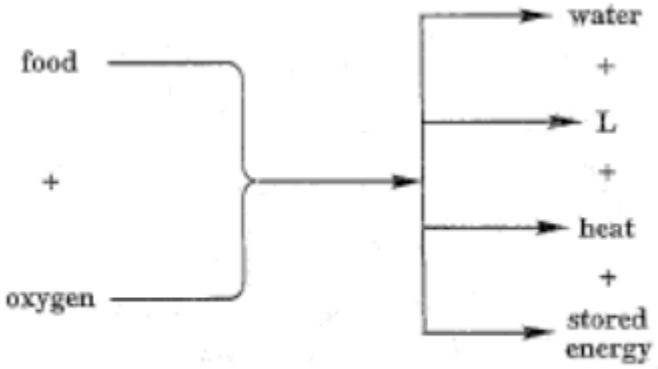

Figure 4

2005-9.

a. Name the product represented by the letter **L**. (1 mark)

b.

(i) What type of respiration is shown in **Figure 4**? (1 mark)

(ii) Give a reason for your answer to b.(i). (1mark)

c. State any **two** ways in which the stored energy may be used. (2 marks)

Section C

2011-15.

During winter, a framer decided to light a charcoal burner to warm calves in a modern cattle khola. Before the charcoal completely got burnt, it was taken into the khola which had its windows closed. The following morning the calves were found dead. In an essay form, explain the steps that led to the death of the calves. (10 marks)

2010-17.

Describe an experiment that could be carried out to investigate the effect of exercise on breathing rate in human beings. Your essay should include procedure, expected results and conclusion. (10 marks)

2006-15.

Describe the breathing mechanism in human beings. Your answer should be in an essay form. (10 marks)

2004-17.

Explain the mechanism of breathing in fish. (10 marks)